

FLIPKART CUSTOMER SEGMENTATION ANALYSIS

USING BUSINESS ANALYTICS

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1. Objective

The primary objective of this project is to establish a comprehensive descriptive analytics framework for evaluating customer purchasing behavior on Flipkart. The study answers several key business questions:

- What percentage of customers are new versus returning?
- Which cities contribute the highest revenue?
- Which product categories generate the most sales?
- How does daily revenue change throughout the month?
- What customer segments contribute the greatest business value?

2. How This Would Work at Real Scale

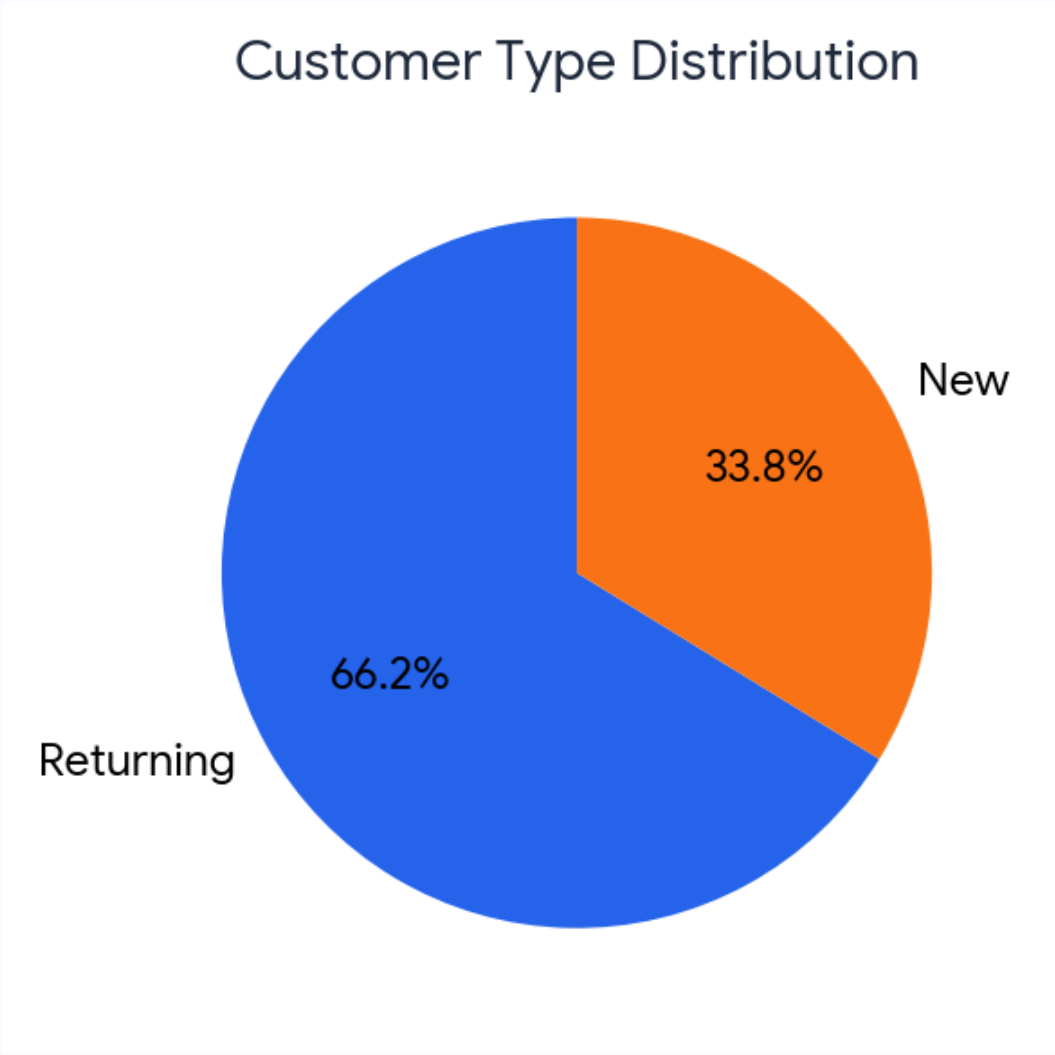
Processing millions of customer transactions every day requires a scalable data analytics architecture before business insights can be generated.

Layer	Purpose	Typical Tools
Data Collection	Capture customer orders from the Flipkart application APIs	REST APIs, Kafka
Data Processing	Clean and organize customer transactions	Python, Pandas
Data Storage	Store transactional and historical customer data	MySQL / Snowflake
Analytics Layer	Generate customer segmentation and KPIs	Python / SQL
Visualization Layer	Display dashboards and reports	Matplotlib / PowerBI

3. Customer Type Distribution

The customer segmentation analysis indicates that **65.6%** of all orders were placed by returning customers, while **34.4%** were placed by new customers. This demonstrates that Flipkart has developed a strong base of loyal customers who repeatedly use the platform for e-commerce purchases.

A high proportion of returning customers generally indicates effective customer retention strategies and positive user experience. However, the presence of new customers also reflects continuous market expansion and successful customer acquisition efforts.

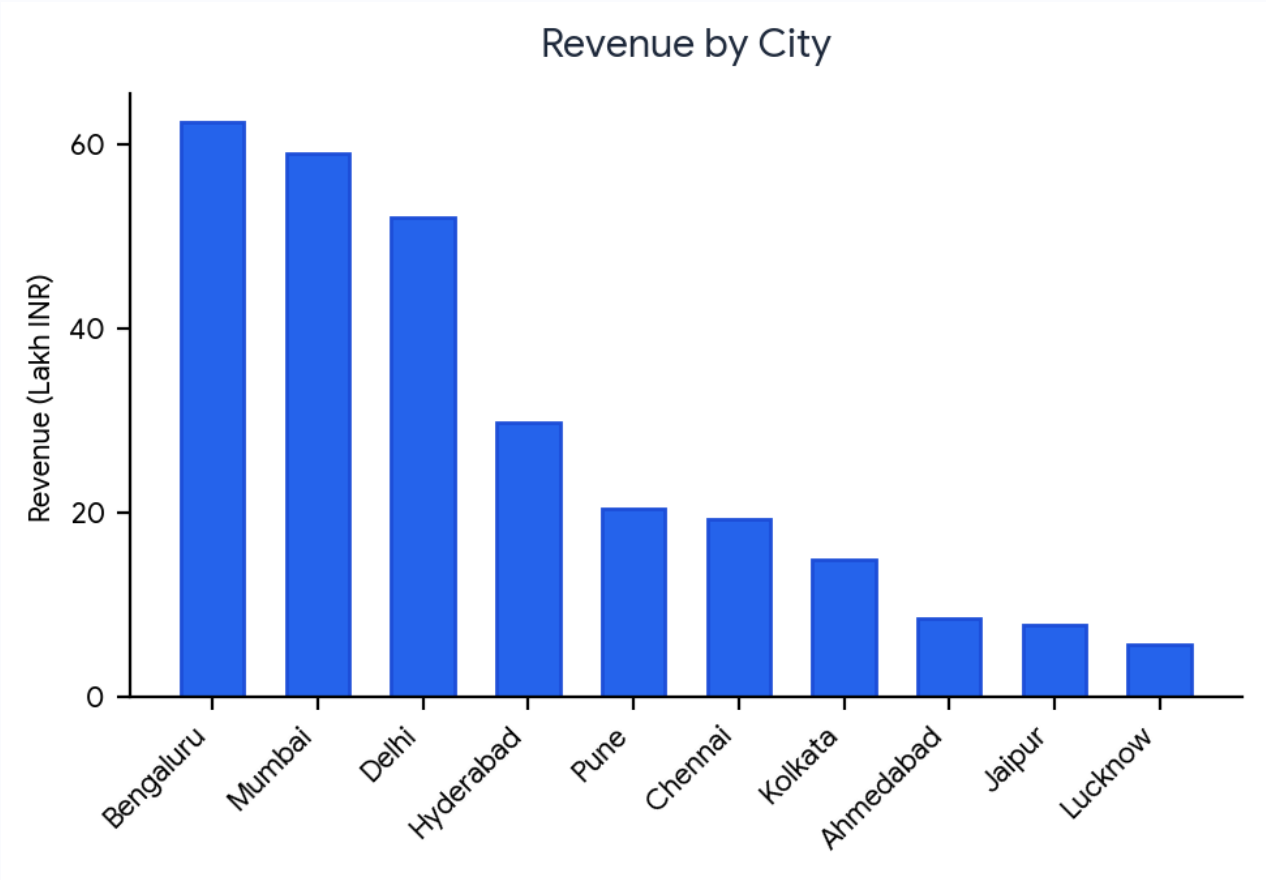


65.6% /
34.4%

RETURNING
VS. NEW
CUSTOMERS

4. Revenue by City

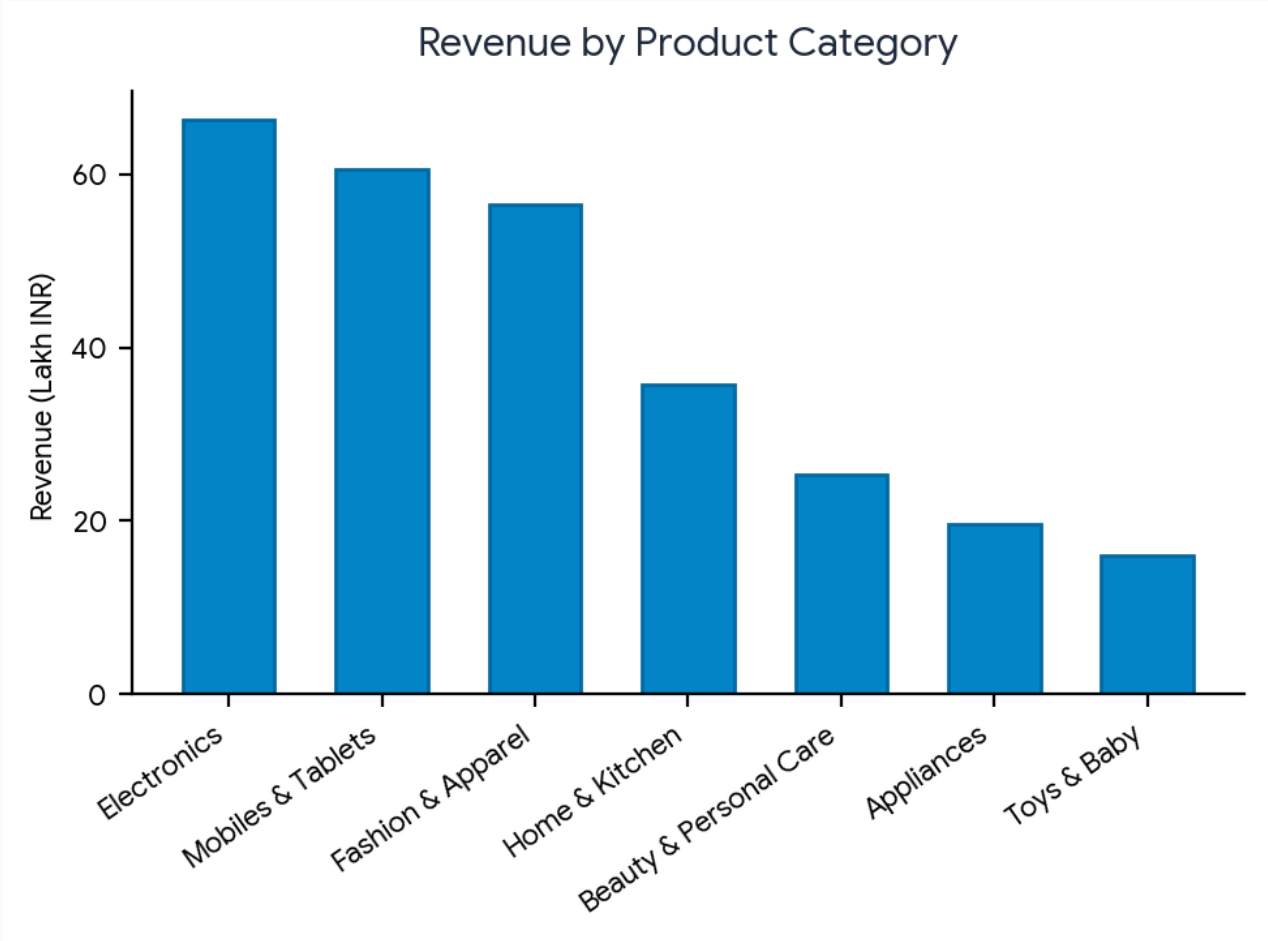
The city-wise revenue analysis reveals that Delhi, Mumbai, and Bengaluru generated the highest revenue during July 2026. These metropolitan cities account for a significant share of overall business revenue due to their higher population density, increased purchasing power, and greater adoption of e-commerce services.



Cities such as Lucknow, Kolkata, and Jaipur generated comparatively lower revenue, suggesting opportunities for targeted promotional campaigns and logistics expansion. This analysis supports decisions related to warehouse allocation, delivery operations, and regional marketing strategies.

5. Revenue by Product Category

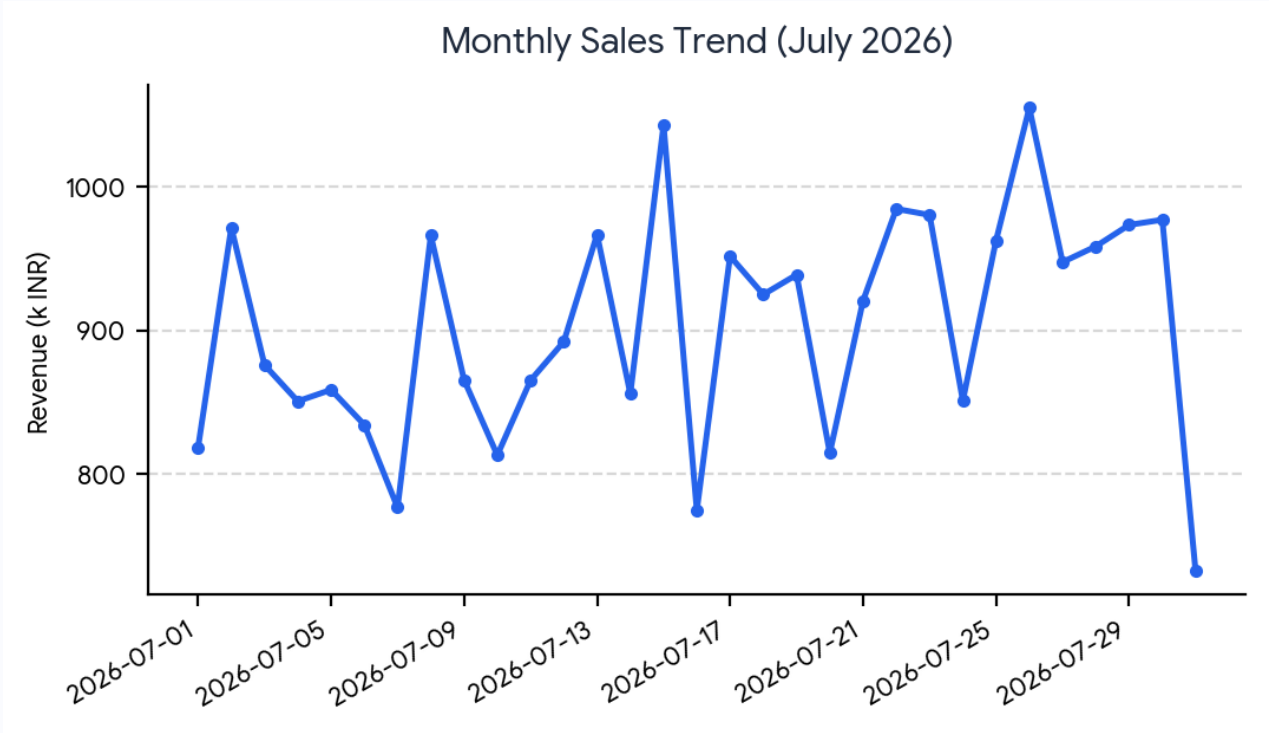
Among all product categories, Electronics and Mobiles & Tablets generated the highest revenue, followed by Fashion & Apparel. These categories represent high-value and frequently purchased consumer items with consistently high market demand.



Lower revenue was observed in categories such as Toys & Baby, indicating comparatively lower order volume during the analysis period. Category-level analysis enables Flipkart to optimize inventory planning, supplier management, and promotional campaigns for high-demand products.

6. Monthly Sales Trend (July 2026)

The daily sales trend demonstrates moderate fluctuations in revenue throughout July 2026. Revenue peaked during the middle of the month and weekend cycles, while a few temporary declines were observed during certain weekdays.



These fluctuations may be influenced by customer demand patterns, promotional campaigns, seasonal purchasing behavior, and order volume variations. Monitoring daily sales trends helps management identify abnormal revenue changes and evaluate the effectiveness of marketing campaigns.

7. Customer Segmentation Analysis

Customers were segmented according to their purchasing behavior and spending patterns. The five primary customer segments include:

- **Premium Customers:** High-frequency shoppers with high average order values.
- **Regular Customers:** Customers who place consistent orders throughout the month.
- **Budget Customers:** Customers primarily purchasing discounted products.
- **New Customers:** First-time users placing their initial orders.
- **Returning Customers:** Existing customers making repeat purchases.

Customer segmentation enables personalized marketing campaigns, targeted discounts, and improved customer retention strategies.

8. Summary of Key Metrics

Metric	Value
Total Orders	10,000
Analysis Period	July 2026
Returning Customers	65.6%
New Customers	34.4%
Highest Revenue City	Delhi / Bengaluru

Metric	Value
Highest Revenue Category	Electronics / Mobiles
Total Cities Analyzed	10
Product Categories	7
Highest Daily Revenue	₹10.5 Lakhs (Approx.)

9. Business Insights & FAQs

Which customer segment generates the highest revenue?

The customer segment generating the highest revenue can be identified by comparing the total order value contributed by each segment. In the dataset, Premium and Regular customers represent a substantial financial value due to higher average order values, making customer retention and premium loyalty programs an important business priority.

Which payment method is most commonly used?

Payment method analysis helps identify how customers prefer to pay for their Flipkart orders. In the July 2026 dataset, **UPI is the most commonly used payment method (48.0%)**. Credit Card is the second-most-used method (17.6%), followed by Debit Card (13.7%), Wallet payments (12.5%), and Cash on Delivery (8.1%).

Descriptive to Predictive Analytics:

The current project focuses primarily on descriptive analytics, which explains what happened in the available sales data. In future, predictive analytics can be applied to forecast customer demand, identify customers likely to stop purchasing (churn prediction), predict peak sales, and estimate fulfillment warehouse inventory requirements.

10. Total Revenue & Key Observations

Total revenue represents the overall sales value generated from recorded orders. It is one of the primary indicators of business performance. In this study, revenue is analyzed across different cities, product categories, and customer segments to understand their contribution to overall sales.

- Returning customers contribute the majority of total order volume.
- Metropolitan cities remain the strongest revenue-generating markets.
- Electronics and Mobile products consistently achieve the highest sales value.
- Daily sales exhibit moderate fluctuations throughout the month.
- Customer segmentation provides valuable insights for personalized marketing and customer retention.